

Case Study 3

Doris Mbitazi Asongafac

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```
# Loading the necessary packages  
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.4.3
```

```
## Registered S3 method overwritten by 'quantmod':  
##   method             from  
##   as.zoo.data.frame zoo
```

```
library(quantmod)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

```
##  
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric
```

```
## Loading required package: TTR
```

```
library(lubridate)
```

```
##  
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':  
##  
##   date, intersect, setdiff, union
```

```
library(readxl)
```

```
# Read in the data sets
dow_jones = read.csv("/Users/doris/OneDrive/Documents/DA6813/dow_jones_index.data")
SP500Data <- read.csv("/Users/doris/OneDrive/Documents/DA6813/Download Data - INDEX_US_S&P US_SPX.csv",

# Remove any missing values if needed
dow_jones = na.omit(dow_jones)
anyNA(dow_jones)
```

```
## [1] FALSE
```

```
SP500Data = na.omit(SP500Data)
anyNA(SP500Data)
```

```
## [1] FALSE
```

```
# See what the data classes are
str(dow_jones)
```

```
## 'data.frame': 720 obs. of 16 variables:
## $ quarter : int 1 1 1 1 1 1 1 1 1 1 ...
## $ stock : chr "AA" "AA" "AA" "AA" ...
## $ date : chr "1/14/2011" "1/21/2011" "1/28/2011" "2/4/2011" ...
## $ open : chr "$16.71" "$16.19" "$15.87" "$16.18" ...
## $ high : chr "$16.71" "$16.38" "$16.63" "$17.39" ...
## $ low : chr "$15.64" "$15.60" "$15.82" "$16.18" ...
## $ close : chr "$15.97" "$15.79" "$16.13" "$17.14" ...
## $ volume : int 242963398 138428495 151379173 154387761 114691279 800238
## $ percent_change_price : num -4.428 -2.471 1.638 5.933 0.231 ...
## $ percent_change_volume_over_last_wk: num 1.38 -43.02 9.36 1.99 -25.71 ...
## $ previous_weeks_volume : int 239655616 242963398 138428495 151379173 154387761 114691
## $ next_weeks_open : chr "$16.19" "$15.87" "$16.18" "$17.33" ...
## $ next_weeks_close : chr "$15.79" "$16.13" "$17.14" "$17.37" ...
## $ percent_change_next_weeks_price : num -2.471 1.638 5.933 0.231 -0.633 ...
## $ days_to_next_dividend : int 19 12 5 97 90 83 76 69 62 55 ...
## $ percent_return_next_dividend : num 0.188 0.19 0.186 0.175 0.173 ...
## - attr(*, "na.action")= 'omit' Named int [1:30] 1 13 25 37 49 61 73 85 97 109 ...
## ..- attr(*, "names")= chr [1:30] "1" "13" "25" "37" ...
```

```
# Convert misclassified columns to the correct format
dow_jones$open=as.numeric(gsub("\\$", "", dow_jones$open))

dow_jones$high=as.numeric(gsub("\\$", "", dow_jones$high))

dow_jones$low=as.numeric(gsub("\\$", "", dow_jones$low))

dow_jones$close=as.numeric(gsub("\\$", "", dow_jones$close))

dow_jones$next_weeks_open=as.numeric(gsub("\\$", "", dow_jones$next_weeks_open))

dow_jones$next_weeks_close=as.numeric(gsub("\\$", "", dow_jones$next_weeks_close))
```

```
dow_jones$date <- mdy(dow_jones$date)
```

```
dow_jones$stock = as.factor(dow_jones$stock)
dow_jones$quarter = as.factor(dow_jones$quarter)
```

```
# Verify the change
str(dow_jones)
```

```
## 'data.frame':    720 obs. of  16 variables:
##  $ quarter          : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ stock             : Factor w/ 30 levels "AA","AXP","BA",...: 1 1 1 1 1 1 1 1 1 1 .
##  $ date              : Date, format: "2011-01-14" "2011-01-21" ...
##  $ open              : num  16.7 16.2 15.9 16.2 17.3 ...
##  $ high              : num  16.7 16.4 16.6 17.4 17.5 ...
##  $ low               : num  15.6 15.6 15.8 16.2 17 ...
##  $ close             : num  16 15.8 16.1 17.1 17.4 ...
##  $ volume            : int  242963398 138428495 151379173 154387761 114691279 800238...
##  $ percent_change_price : num  -4.428 -2.471 1.638 5.933 0.231 ...
##  $ percent_change_volume_over_last_wk: num  1.38 -43.02 9.36 1.99 -25.71 ...
##  $ previous_weeks_volume : int  239655616 242963398 138428495 151379173 154387761 114691...
##  $ next_weeks_open     : num  16.2 15.9 16.2 17.3 17.4 ...
##  $ next_weeks_close    : num  15.8 16.1 17.1 17.4 17.3 ...
##  $ percent_change_next_weeks_price : num  -2.471 1.638 5.933 0.231 -0.633 ...
##  $ days_to_next_dividend : int  19 12 5 97 90 83 76 69 62 55 ...
##  $ percent_return_next_dividend : num  0.188 0.19 0.186 0.175 0.173 ...
##  - attr(*, "na.action")= 'omit' Named int [1:30] 1 13 25 37 49 61 73 85 97 109 ...
##  ..- attr(*, "names")= chr [1:30] "1" "13" "25" "37" ...
```

```
str(SP500Data)
```

```
## 'data.frame':    24 obs. of  5 variables:
##  $ Date : chr  "06/24/2011" "06/17/2011" "06/10/2011" "06/03/2011" ...
##  $ Open : chr  "1,271.50" "1,271.31" "1,300.26" "1,331.10" ...
##  $ High : chr  "1,298.61" "1,292.50" "1,300.26" "1,345.20" ...
##  $ Low  : chr  "1,262.87" "1,258.07" "1,268.28" "1,297.90" ...
##  $ Close: chr  "1,268.45" "1,271.50" "1,270.98" "1,300.16" ...
```

```
#Rename for consistency
```

```
SP500Data$Date <- as.Date(SP500Data$Date, format="%m/%d/%Y")
SP500Data$Open <- as.numeric(gsub(",", "", SP500Data$Open))
SP500Data$High <- as.numeric(gsub(",", "", SP500Data$High))
SP500Data$Low <- as.numeric(gsub(",", "", SP500Data$Low))
SP500Data$Close <- as.numeric(gsub(",", "", SP500Data$Close))
```

```
str(SP500Data)
```

```
## 'data.frame':    24 obs. of  5 variables:
##  $ Date : Date, format: "2011-06-24" "2011-06-17" ...
##  $ Open : num  1272 1271 1300 1331 1333 ...
##  $ High : num  1299 1292 1300 1345 1335 ...
##  $ Low  : num  1263 1258 1268 1298 1312 ...
##  $ Close: num  1268 1272 1271 1300 1331 ...
```

```
colnames(SP500Data)[colnames(SP500Data) == "Date"] <- "date"
```

```
summary(SP500Data)
```

```
##      date      Open      High      Low
## Min.   :2011-01-14  Min.   :1271  Min.   :1292  Min.   :1249
## 1st Qu.:2011-02-23  1st Qu.:1291  1st Qu.:1302  1st Qu.:1276
## Median :2011-04-04  Median :1319  Median :1333  Median :1296
## Mean   :2011-04-04  Mean   :1313  Mean   :1327  Mean   :1296
## 3rd Qu.:2011-05-14  3rd Qu.:1333  3rd Qu.:1341  3rd Qu.:1313
## Max.   :2011-06-24  Max.   :1365  Max.   :1371  Max.   :1332
##      Close
## Min.   :1268
## 1st Qu.:1291
## Median :1320
## Mean   :1313
## 3rd Qu.:1333
## Max.   :1364
```

```
summary(dow_jones)
```

```
## quarter  stock      date      open      high
## 1:330    AA      : 24  Min.   :2011-01-14  Min.   : 10.59  Min.   : 10.94
## 2:390    AXP      : 24  1st Qu.:2011-02-23  1st Qu.: 29.96  1st Qu.: 30.66
##          BA      : 24  Median :2011-04-04  Median : 46.02  Median : 46.91
##          BAC      : 24  Mean   :2011-04-04  Mean   : 53.75  Mean   : 54.76
##          CAT      : 24  3rd Qu.:2011-05-14  3rd Qu.: 72.81  3rd Qu.: 74.34
##          CSCO     : 24  Max.   :2011-06-24  Max.   :172.11  Max.   :173.54
##          (Other):576
##      low      close      volume      percent_change_price
## Min.   : 10.40  Min.   : 10.52  Min.   :9.719e+06  Min.   : -15.42290
## 1st Qu.: 28.71  1st Qu.: 30.41  1st Qu.:3.069e+07  1st Qu.: -1.29736
## Median : 44.95  Median : 46.12  Median :5.275e+07  Median :  0.00000
## Mean   : 52.73  Mean   : 53.83  Mean   :1.156e+08  Mean   :  0.03014
## 3rd Qu.: 71.11  3rd Qu.: 72.77  3rd Qu.:1.312e+08  3rd Qu.:  1.63087
## Max.   :167.82  Max.   :170.58  Max.   :1.054e+09  Max.   :  9.88223
##
## percent_change_volume_over_last_wk previous_weeks_volume next_weeks_open
## Min.   : -61.4332  Min.   :9.719e+06  Min.   : 10.52
## 1st Qu.: -19.8043  1st Qu.:3.068e+07  1st Qu.: 30.39
## Median :  0.5126  Median :5.295e+07  Median : 46.04
## Mean   :  5.5936  Mean   :1.174e+08  Mean   : 53.81
## 3rd Qu.: 21.8006  3rd Qu.:1.333e+08  3rd Qu.: 72.81
## Max.   :327.4089  Max.   :1.453e+09  Max.   :172.11
##
## next_weeks_close percent_change_next_weeks_price days_to_next_dividend
## Min.   : 10.52  Min.   : -15.4229  Min.   :  0.00
## 1st Qu.: 30.51  1st Qu.: -1.2524  1st Qu.: 24.00
## Median : 45.93  Median :  0.0360  Median : 47.00
## Mean   : 53.97  Mean   :  0.1933  Mean   : 52.26
## 3rd Qu.: 73.00  3rd Qu.:  1.7927  3rd Qu.: 69.25
## Max.   :174.54  Max.   :  9.8822  Max.   :329.00
```

```
##
## percent_return_next_dividend
## Min. :0.06557
## 1st Qu.:0.53409
## Median :0.68014
## Mean :0.69156
## 3rd Qu.:0.85402
## Max. :1.56421
##
```

```
Dow = split(dow_jones, dow_jones$stock)
```

```
# Compute the returns and remove any missing values.
```

```
ReturnAA = na.omit(Delt(Dow$AA[,7]))
ReturnAXP = na.omit(Delt(Dow$AXP[,7]))
ReturnBA = na.omit(Delt(Dow$BA[,7]))
ReturnBAC = na.omit(Delt(Dow$BAC[,7]))
ReturnCAT = na.omit(Delt(Dow$CAT[,7]))
ReturnCSCO = na.omit(Delt(Dow$CSCO[,7]))
ReturnCVX = na.omit(Delt(Dow$CVX[,7]))
ReturnDD = na.omit(Delt(Dow$DD[,7]))
ReturnDIS = na.omit(Delt(Dow$DIS[,7]))
ReturnGE = na.omit(Delt(Dow$GE[,7]))
ReturnHD = na.omit(Delt(Dow$HD[,7]))
ReturnHPQ = na.omit(Delt(Dow$HPQ[,7]))
ReturnIBM = na.omit(Delt(Dow$IBM[,7]))
ReturnINTC = na.omit(Delt(Dow$INTC[,7]))
ReturnJNJ = na.omit(Delt(Dow$JNJ[,7]))
ReturnJPM = na.omit(Delt(Dow$JPM[,7]))
ReturnKO = na.omit(Delt(Dow$KO[,7]))
ReturnKRFT = na.omit(Delt(Dow$KRFT[,7]))
ReturnMCD = na.omit(Delt(Dow$MCD[,7]))
ReturnMMM = na.omit(Delt(Dow$MMM[,7]))
ReturnMRK = na.omit(Delt(Dow$MRK[,7]))
ReturnMSFT = na.omit(Delt(Dow$MSFT[,7]))
ReturnPFE = na.omit(Delt(Dow$PFE[,7]))
ReturnPG = na.omit(Delt(Dow$PG[,7]))
ReturnT = na.omit(Delt(Dow$T[,7]))
ReturnTRV = na.omit(Delt(Dow$TRV[,7]))
ReturnUTX = na.omit(Delt(Dow$UTX[,7]))
ReturnVZ = na.omit(Delt(Dow$VZ[,7]))
ReturnWMT = na.omit(Delt(Dow$WMT[,7]))
ReturnXOM = na.omit(Delt(Dow$XOM[,7]))
ReturnSP500 = na.omit(Delt(SP500Data[,5]))
```

```
# Merge the returns of all stocks into single data
```

```
MyData = cbind(ReturnSP500,
               ReturnAA,
               ReturnAXP,
               ReturnBA,
               ReturnBAC,
               ReturnCAT,
               ReturnCSCO,
```

```

ReturnCVX,
ReturnDD,
ReturnDIS,
ReturnGE,
ReturnHD,
ReturnHPQ,
ReturnIBM,
ReturnINTC,
ReturnJNJ,
ReturnJPM,
ReturnKO,
ReturnKRFT,
ReturnMCD,
ReturnMMM,
ReturnMRK,
ReturnMSFT,
ReturnPFE,
ReturnPG,
ReturnT,
ReturnTRV,
ReturnUTX,
ReturnVZ,
ReturnWMT,
ReturnXOM)

```

Weekly Return of Investment of Interest

```

colnames(MyData) = c("SP500", "AA", "AXP", "BA", "BAC", "CAT", "CSCO", "CVX", "DD", "DIS", "GE", "HD",
                     "VZ", "WMT", "XOM")

```

```

head(MyData)

```

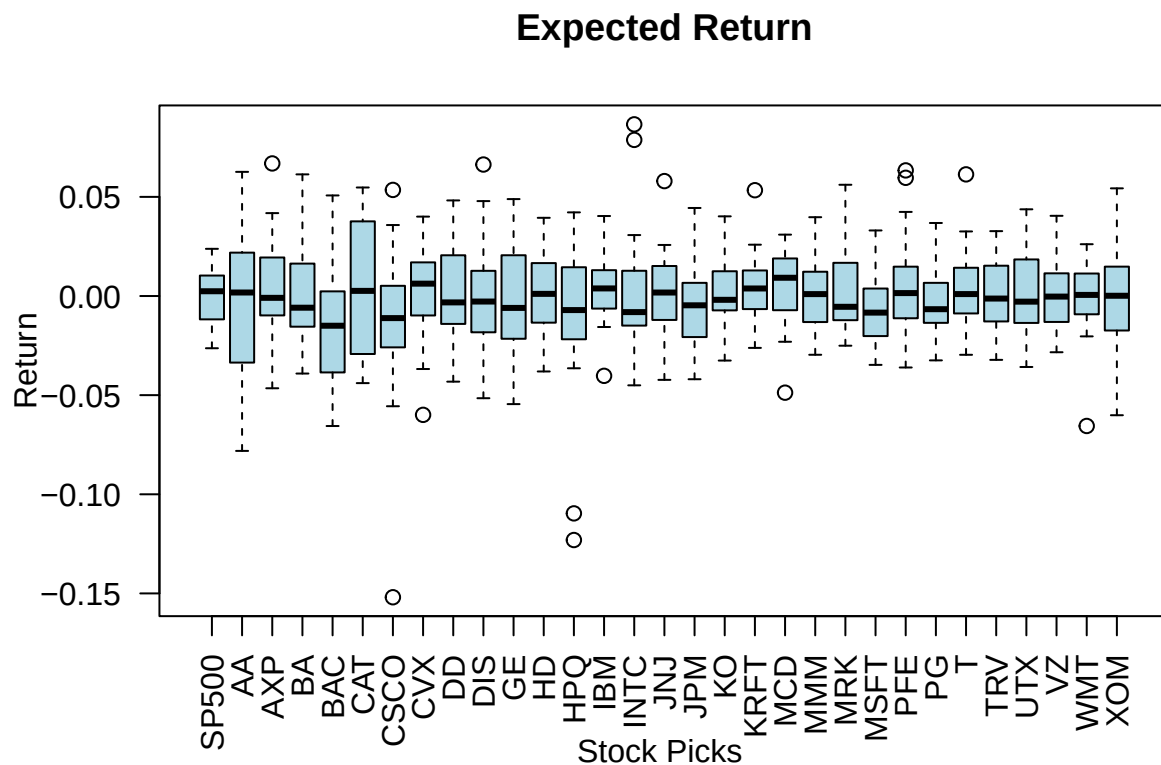
```

##           SP500           AA           AXP           BA           BAC
## [1,]  0.0024045094 -0.011271133 -0.0054054054  0.02297702 -0.065573770
## [2,] -0.0004089658  0.021532616 -0.0465217391 -0.03417969 -0.045614035
## [3,]  0.0229586618  0.062616243 -0.0009119927  0.03105590  0.050735294
## [4,]  0.0237970711  0.013418903  0.0668644455  0.01064724  0.033589923
## [5,]  0.0016302306 -0.005181347 -0.0260962567  0.01247574 -0.001354096
## [6,]  0.0033751603 -0.034722222 -0.0439270810 -0.01013143 -0.037288136
##           CAT           CSCO           CVX           DD           DIS           GE
## [1,] -0.01340283 -0.02310231  0.010233761 -0.02911647  0.011453296  0.048884166
## [2,]  0.03159030  0.01013514 -0.004371934  0.04012410 -0.022395571  0.023302938
## [3,]  0.04086538  0.05351171  0.040055692  0.04454166  0.047876448  0.017821782
## [4,]  0.03966262 -0.15192744 -0.006796416  0.03902532  0.066322771  0.037451362
## [5,]  0.02240680  0.00802139  0.023535511  0.02565042  0.003455425  0.005157056
## [6,] -0.03646325 -0.01114058  0.034238250 -0.03411933 -0.014003673 -0.028917910
##           HD           HPQ           IBM           INTC           JNJ
## [1,]  0.017275007  0.0211891892  0.0366666667 -0.012333966  0.001758593
## [2,]  0.005204054 -0.0364175312  0.0238585209  0.030739673 -0.042291733
## [3,]  0.002724796  0.0421885300  0.0300860499  0.010251631  0.013831028
## [4,]  0.018478261  0.0255112798 -0.0009146341  0.003690037 -0.002301118
## [5,]  0.026680896  0.0006167763  0.0060421117  0.017463235  0.006754530
## [6,] -0.036382536 -0.1230737621 -0.0155302111 -0.012646793 -0.024054983
##           JPM           KO           KRFT           MCD           MMM
## [1,]  0.008461367 -0.005702519  0.000319081  0.012827437  0.013507378

```

```
## [2,] -0.016559947 -0.008921459 -0.026156300 -0.023063592 -0.020719005
## [3,]  0.001122586  0.005626105  0.021618081  0.010507642  0.009720952
## [4,]  0.044404575  0.016144501 -0.016992626  0.028224173  0.039755352
## [5,]  0.030706463  0.015416077  0.008153947 -0.000131337  0.012636166
## [6,] -0.027500000 -0.003718048  0.025881592 -0.022198870 -0.029152324
##           MRK           MSFT           PFE           PG           T
## [1,] -0.009640666 -0.0098939929  0.001090513  0.005798871 -0.003517411
## [2,] -0.024483776 -0.0096359743 -0.011437908 -0.025944470 -0.029650547
## [3,] -0.005443000  0.0007207207  0.063360882 -0.009190031  0.017460895
## [4,]  0.005472788 -0.0187252431 -0.024352332  0.017607294  0.017876296
## [5,] -0.006652555 -0.0069724771  0.019118428 -0.006642979  0.003512469
## [6,] -0.020091324 -0.0188470067 -0.017196456 -0.022706065 -0.015400770
##           TRV           UTX           VZ           WMT           XOM
## [1,]  0.006772835  0.014162873 -0.014382403  0.016785258  0.0146454265
## [2,]  0.014727273  0.015336658  0.019456366  0.017405347  0.0001266143
## [3,]  0.028668697  0.013385730  0.019085041 -0.011816578  0.0543106722
## [4,]  0.027521338  0.032476975  0.002203250 -0.006068178 -0.0055235351
## [5,]  0.032717410 -0.002230047  0.006320418 -0.005566529  0.0202849553
## [6,] -0.021667761 -0.019291848 -0.017749863 -0.065547129  0.0099408284
```

```
boxplot(MyData,
  main = "Expected Return",
  xlab = "Stock Picks ",
  ylab = "Return",
  col = "lightblue",
  las = 2) # Rotate labels for readability
```



```

# Compute mean and stdev for the returns.
DataMean=apply(MyData, 2, mean)
DataSD=apply(MyData, 2, sd)

# Take a look at the means and standard deviations.
cbind(DataMean,DataSD)

```

```

##           DataMean      DataSD
## SP500  0.0009469730 0.01480999
## AA     -0.0014758553 0.03486277
## AXP     0.0022992844 0.02809141
## BA      0.0011022005 0.02795967
## BAC     -0.0156049349 0.02911342
## CAT     0.0032480226 0.03411175
## CSC0    -0.0143473271 0.03971904
## CVX     0.0026118394 0.02482674
## DD      0.0022017080 0.02723925
## DIS     -0.0015588482 0.02808084
## GE      -0.0016902251 0.02574530
## HD      -0.0007599981 0.02198521
## HPQ     -0.0113893855 0.03927282
## IBM     0.0043318296 0.01836236
## INTC    0.0007425153 0.03258276
## JNJ     0.0019245403 0.02114399
## JPM     -0.0053481151 0.02181402
## KO      0.0013550453 0.01664583
## KRFT    0.0044681344 0.01816930
## MCD     0.0045364377 0.01954457
## MMM     0.0015614511 0.02073424
## MRK     0.0006072965 0.02069553
## MSFT    -0.0064307236 0.01898709
## PFE     0.0042807309 0.02655822
## PG      -0.0018427834 0.01779427
## T       0.0031507287 0.01934575
## TRV     0.0017787893 0.01920565
## UTX     0.0030140170 0.02130799
## VZ      0.0008114707 0.01799993
## WMT     -0.0017666125 0.01910305
## XOM     -0.0002927384 0.02512918

```

```

# Using LM here as interpretation
# is key.
# Linear Regression models

# Model AA
lm.AA <- lm(AA ~ SP500, data = as.data.frame(MyData))
summary(lm.AA)

```

```

##
## Call:
## lm(formula = AA ~ SP500, data = as.data.frame(MyData))
##
## Residuals:

```



```
##           Min           1Q       Median           3Q           Max
## -0.072080 -0.029183 -0.000855  0.022727  0.066652
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.001766   0.007393  -0.239   0.814
## SP500        0.306355   0.509317   0.602   0.554
##
## Residual standard error: 0.03538 on 21 degrees of freedom
## Multiple R-squared:  0.01694, Adjusted R-squared:  -0.02988
## F-statistic: 0.3618 on 1 and 21 DF, p-value: 0.5539
```

```
BetaAA <- summary(lm.AA)$coefficients[2, 1]
```

```
# Model AXP
```

```
lm.AXP <- lm(AXP ~ SP500, data = as.data.frame(MyData))
summary(lm.AXP)
```

```
##
## Call:
## lm(formula = AXP ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##           Min           1Q       Median           3Q           Max
## -0.048172 -0.013327  0.001369  0.013786  0.046256
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.001540   0.005446   0.283   0.7800
## SP500        0.801279   0.375167   2.136   0.0446 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02606 on 21 degrees of freedom
## Multiple R-squared:  0.1785, Adjusted R-squared:  0.1393
## F-statistic: 4.562 on 1 and 21 DF, p-value: 0.04463
```

```
BetaAXP <- summary(lm.AXP)$coefficients[2, 1]
```

```
#Model BA
```

```
lm.BA <- lm(BA ~ SP500, data = as.data.frame(MyData))
summary(lm.BA)
```

```
##
## Call:
## lm(formula = BA ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##           Min           1Q       Median           3Q           Max
## -0.042628 -0.016080 -0.007602  0.014311  0.065337
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.0007631  0.0058713   0.130   0.898
## SP500       0.3581406  0.4044911   0.885   0.386
##
## Residual standard error: 0.0281 on 21 degrees of freedom
## Multiple R-squared:  0.03599,    Adjusted R-squared:  -0.009918
## F-statistic: 0.784 on 1 and 21 DF,  p-value: 0.386
```

```
BetaBA <- summary(lm.BA)$coefficients[2, 1]
```

```
#Model BAC
```

```
lm.BAC <- lm(BAC ~ SP500, data = as.data.frame(MyData))
summary(lm.BAC)
```

```
##
## Call:
## lm(formula = BAC ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.051800 -0.013741  0.002985  0.013326  0.039262
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.016795   0.004788  -3.507  0.00210 **
## SP500        1.256567   0.329891   3.809  0.00102 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02292 on 21 degrees of freedom
## Multiple R-squared:  0.4086, Adjusted R-squared:  0.3804
## F-statistic: 14.51 on 1 and 21 DF,  p-value: 0.001025
```

```
BetaBAC <- summary(lm.BAC)$coefficients[2, 1]
```

```
#Model for CAT
```

```
lm.CAT <- lm(CAT ~ SP500, data = as.data.frame(MyData))
summary(lm.CAT)
```

```
##
## Call:
## lm(formula = CAT ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.051372 -0.032386  0.003282  0.029367  0.054203
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.002917   0.007211   0.404   0.690
## SP500       0.349837   0.496789   0.704   0.489
##
```

```
## Residual standard error: 0.03451 on 21 degrees of freedom
## Multiple R-squared:  0.02307,    Adjusted R-squared:  -0.02345
## F-statistic: 0.4959 on 1 and 21 DF,  p-value: 0.489
```

```
BetaCAT <- summary(lm.CAT)$coefficients[2, 1]
```

```
#Model for CSCO
```

```
lm.CSCO <- lm(CSCO ~ SP500, data = as.data.frame(MyData))
summary(lm.CSCO)
```

```
##
## Call:
## lm(formula = CSCO ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.142264	-0.011737	0.002709	0.018442	0.063347

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.01454	0.00847	-1.717	0.101
SP500	0.20498	0.58353	0.351	0.729

```
##
## Residual standard error: 0.04053 on 21 degrees of freedom
## Multiple R-squared:  0.005841,    Adjusted R-squared:  -0.0415
## F-statistic: 0.1234 on 1 and 21 DF,  p-value: 0.7289
```

```
BetaCSCO <- summary(lm.CSCO)$coefficients[2, 1]
```

```
#Model for CVX
```

```
lm.CVX <- lm(CVX ~ SP500, data = as.data.frame(MyData))
summary(lm.CVX)
```

```
##
## Call:
## lm(formula = CVX ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.060691	-0.011020	0.001877	0.013173	0.040863

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.002759	0.005287	0.522	0.607
SP500	-0.155348	0.364236	-0.427	0.674

```
##
## Residual standard error: 0.0253 on 21 degrees of freedom
## Multiple R-squared:  0.008588,    Adjusted R-squared:  -0.03862
## F-statistic: 0.1819 on 1 and 21 DF,  p-value: 0.6741
```

```
BetaCVX <- summary(lm.CVX)$coefficients[2, 1]
```

```
#Model for DD
```

```
lm.DD <- lm(DD ~ SP500, data = as.data.frame(MyData))  
summary(lm.DD)
```

```
##  
## Call:  
## lm(formula = DD ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.045028 -0.019426 -0.001111  0.023795  0.043311   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.001823   0.005687   0.321   0.752      
## SP500        0.399690   0.391765   1.020   0.319      
##  
## Residual standard error: 0.02721 on 21 degrees of freedom  
## Multiple R-squared:  0.04722,    Adjusted R-squared:  0.001854   
## F-statistic: 1.041 on 1 and 21 DF,  p-value: 0.3192
```

```
BetaDD <- summary(lm.DD)$coefficients[2, 1]
```

```
#Model for DIS
```

```
lm.DIS <- lm(DIS ~ SP500, data = as.data.frame(MyData))  
summary(lm.DIS)
```

```
##  
## Call:  
## lm(formula = DIS ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.038637 -0.017770 -0.005468  0.011312  0.054707   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.002290   0.005486  -0.417   0.6806      
## SP500        0.771826   0.377926   2.042   0.0539 .      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.02625 on 21 degrees of freedom  
## Multiple R-squared:  0.1657, Adjusted R-squared:  0.126   
## F-statistic: 4.171 on 1 and 21 DF,  p-value: 0.05388
```

```
BetaDIS <- summary(lm.DIS)$coefficients[2, 1]
```

```
#Model for GE
```

```
lm.GE <- lm(GE ~ SP500, data = as.data.frame(MyData))  
summary(lm.GE)
```

```
##  
## Call:  
## lm(formula = GE ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.038389 -0.020116  0.000577  0.015061  0.049532   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.002368   0.005018  -0.472   0.642      
## SP500         0.715414   0.345731   2.069   0.051 .      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.02402 on 21 degrees of freedom  
## Multiple R-squared:  0.1694, Adjusted R-squared:  0.1298   
## F-statistic: 4.282 on 1 and 21 DF,  p-value: 0.05105
```

```
BetaGE <- summary(lm.GE)$coefficients[2, 1]
```

```
#Model for HD
```

```
lm.HD <- lm(HD ~ SP500, data = as.data.frame(MyData))  
summary(lm.HD)
```

```
##  
## Call:  
## lm(formula = HD ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.036701 -0.014744  0.002064  0.011387  0.046503   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.001181   0.004487  -0.263   0.795      
## SP500         0.444162   0.309101   1.437   0.165      
##  
## Residual standard error: 0.02147 on 21 degrees of freedom  
## Multiple R-squared:  0.08952, Adjusted R-squared:  0.04617   
## F-statistic: 2.065 on 1 and 21 DF,  p-value: 0.1655
```

```
BetaHD <- summary(lm.HD)$coefficients[2, 1]
```

```
#Model for HPQ
```

```
lm.HPQ <- lm(HPQ ~ SP500, data = as.data.frame(MyData))  
summary(lm.HPQ)
```

```
##
## Call:
## lm(formula = HPQ ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.111113 -0.010222  0.005085  0.024105  0.058758
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.011167   0.008366  -1.335   0.196
## SP500        -0.235339   0.576382  -0.408   0.687
##
## Residual standard error: 0.04004 on 21 degrees of freedom
## Multiple R-squared:  0.007876,    Adjusted R-squared:  -0.03937
## F-statistic: 0.1667 on 1 and 21 DF,  p-value: 0.6872
```

```
BetaHPQ <- summary(lm.HPQ)$coefficients[2, 1]
```

```
#Model for IBM
lm.IBM <- lm(IBM ~ SP500, data = as.data.frame(MyData))
summary(lm.IBM)
```

```
##
## Call:
## lm(formula = IBM ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.038869 -0.009825 -0.002418  0.010032  0.040041
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.004063   0.003823   1.063   0.300
## SP500        0.283707   0.263382   1.077   0.294
##
## Residual standard error: 0.0183 on 21 degrees of freedom
## Multiple R-squared:  0.05236,    Adjusted R-squared:  0.007233
## F-statistic:  1.16 on 1 and 21 DF,  p-value: 0.2936
```

```
BetaIBM <- summary(lm.IBM)$coefficients[2, 1]
```

```
#Model for INTC
lm.INTC <- lm(INTC ~ SP500, data = as.data.frame(MyData))
summary(lm.INTC)
```

```
##
## Call:
## lm(formula = INTC ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.04340 -0.01412 -0.01017  0.01056  0.08905
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.000631   0.006959   0.091   0.929
## SP500       0.117811   0.479402   0.246   0.808
##
## Residual standard error: 0.0333 on 21 degrees of freedom
## Multiple R-squared:  0.002868, Adjusted R-squared: -0.04461
## F-statistic: 0.06039 on 1 and 21 DF, p-value: 0.8083
```

```
BetaINTC <- summary(lm.INTC)$coefficients[2, 1]
```

```
#Model for JNJ
```

```
lm.JNJ <- lm(JNJ ~ SP500, data = as.data.frame(MyData))
summary(lm.JNJ)
```

```
##
## Call:
## lm(formula = JNJ ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.04468 -0.01417  0.00033  0.01109  0.04674
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.002247   0.004392   0.512   0.614
## SP500       -0.340509   0.302555  -1.125   0.273
##
## Residual standard error: 0.02102 on 21 degrees of freedom
## Multiple R-squared:  0.05688, Adjusted R-squared:  0.01197
## F-statistic: 1.267 on 1 and 21 DF, p-value: 0.2731
```

```
BetaJNJ <- summary(lm.JNJ)$coefficients[2, 1]
```

```
#Model for JPM
```

```
lm.JPM <- lm(JPM ~ SP500, data = as.data.frame(MyData))
summary(lm.JPM)
```

```
##
## Call:
## lm(formula = JPM ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.035508 -0.014586  0.000598  0.015202  0.036498
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.005897   0.004289  -1.375   0.184
## SP500       0.580051   0.295446   1.963   0.063 .
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02052 on 21 degrees of freedom
## Multiple R-squared:  0.1551, Adjusted R-squared:  0.1149
## F-statistic: 3.855 on 1 and 21 DF,  p-value: 0.06299
```

```
BetaJPM <- summary(lm.JPM)$coefficients[2, 1]
```

```
#Model for KO
```

```
lm.KO <- lm(KO ~ SP500, data = as.data.frame(MyData))
summary(lm.KO)
```

```
##
## Call:
## lm(formula = KO ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.030370 -0.010602 -0.001005  0.011521  0.041325
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.001189   0.003517   0.338   0.739
## SP500        0.175481   0.242260   0.724   0.477
##
## Residual standard error: 0.01683 on 21 degrees of freedom
## Multiple R-squared:  0.02438,    Adjusted R-squared:  -0.02208
## F-statistic: 0.5247 on 1 and 21 DF,  p-value: 0.4768
```

```
BetaKO <- summary(lm.KO)$coefficients[2, 1]
```

```
#Model for KRFT
```

```
lm.KRFT <- lm(KRFT ~ SP500, data = as.data.frame(MyData))
summary(lm.KO)
```

```
##
## Call:
## lm(formula = KO ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.030370 -0.010602 -0.001005  0.011521  0.041325
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.001189   0.003517   0.338   0.739
## SP500        0.175481   0.242260   0.724   0.477
##
## Residual standard error: 0.01683 on 21 degrees of freedom
## Multiple R-squared:  0.02438,    Adjusted R-squared:  -0.02208
## F-statistic: 0.5247 on 1 and 21 DF,  p-value: 0.4768
```



```
BetaKRFT <- summary(lm.KRFT)$coefficients[2, 1]
```

```
#Model for MCD
```

```
lm.MCD <- lm(MCD ~ SP500, data = as.data.frame(MyData))  
summary(lm.MCD)
```

```
##  
## Call:  
## lm(formula = MCD ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.042231 -0.006555  0.002294  0.009673  0.034190   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.004018   0.003804   1.056   0.3028      
## SP500        0.547377   0.262039   2.089   0.0491 *      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.0182 on 21 degrees of freedom  
## Multiple R-squared:  0.172, Adjusted R-squared:  0.1326   
## F-statistic: 4.364 on 1 and 21 DF, p-value: 0.04907
```

```
BetaMCD <- summary(lm.MCD)$coefficients[2, 1]
```

```
#Model for MMM
```

```
lm.MMM <- lm(MMM ~ SP500, data = as.data.frame(MyData))  
summary(lm.MMM)
```

```
##  
## Call:  
## lm(formula = MMM ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.031615 -0.017256  0.002137  0.011217  0.039293   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.001302   0.004349   0.299   0.768      
## SP500        0.273591   0.299618   0.913   0.372      
##  
## Residual standard error: 0.02081 on 21 degrees of freedom  
## Multiple R-squared:  0.03819, Adjusted R-squared: -0.007612   
## F-statistic: 0.8338 on 1 and 21 DF, p-value: 0.3715
```

```
BetaMMM <- summary(lm.MMM)$coefficients[2, 1]
```

```
#Model for MRK
```

```
lm.MRK <- lm(MRK ~ SP500, data = as.data.frame(MyData))  
summary(lm.MRK)
```

```
##  
## Call:  
## lm(formula = MRK ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.026928 -0.015688 -0.004038  0.014662  0.048367   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.0002451  0.0042573   0.058   0.955      
## SP500         0.3824517  0.2932955   1.304   0.206      
##  
## Residual standard error: 0.02037 on 21 degrees of freedom  
## Multiple R-squared:  0.0749, Adjusted R-squared:  0.03085   
## F-statistic: 1.7 on 1 and 21 DF,  p-value: 0.2064
```

```
BetaMRK <- summary(lm.MRK)$coefficients[2, 1]
```

```
#Model for MSFT
```

```
lm.MSFT <- lm(MSFT ~ SP500, data = as.data.frame(MyData))  
summary(lm.MSFT)
```

```
##  
## Call:  
## lm(formula = MSFT ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.026672 -0.015547 -0.000618  0.011350  0.041069   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.006536  0.004046  -1.616   0.121      
## SP500         0.110945  0.278716   0.398   0.695      
##  
## Residual standard error: 0.01936 on 21 degrees of freedom  
## Multiple R-squared:  0.007489, Adjusted R-squared: -0.03977   
## F-statistic: 0.1585 on 1 and 21 DF,  p-value: 0.6946
```

```
BetaMSFT <- summary(lm.MSFT)$coefficients[2, 1]
```

```
#Model for PFE
```

```
lm.PFE <- lm(PFE ~ SP500, data = as.data.frame(MyData))  
summary(lm.PFE)
```

```
##
```

```
## Call:
## lm(formula = PFE ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.039599 -0.022258 -0.001526  0.013618  0.048516
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003826   0.005473   0.699   0.492
## SP500        0.479915   0.377049   1.273   0.217
##
## Residual standard error: 0.02619 on 21 degrees of freedom
## Multiple R-squared:  0.07162,    Adjusted R-squared:  0.02741
## F-statistic:  1.62 on 1 and 21 DF,  p-value: 0.217
```

```
BetaPFE <- summary(lm.PFE)$coefficients[2, 1]
```

```
#Model for PG
lm.PG <- lm(PG ~ SP500, data = as.data.frame(MyData))
summary(lm.PG)
```

```
##
## Call:
## lm(formula = PG ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.032400 -0.011244 -0.002129  0.008728  0.042603
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.002092   0.003713  -0.563   0.579
## SP500        0.263308   0.255816   1.029   0.315
##
## Residual standard error: 0.01777 on 21 degrees of freedom
## Multiple R-squared:  0.04803,    Adjusted R-squared:  0.002694
## F-statistic: 1.059 on 1 and 21 DF,  p-value: 0.3151
```

```
BetaPG <- summary(lm.PG)$coefficients[2, 1]
```

```
#Model for T
lm.T <- lm(T ~ SP500, data = as.data.frame(MyData))
summary(lm.T)
```

```
##
## Call:
## lm(formula = T ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.032182 -0.011611  0.000050  0.004279  0.055697
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.002718   0.003877   0.701   0.491
## SP500       0.456481   0.267079   1.709   0.102
##
## Residual standard error: 0.01855 on 21 degrees of freedom
## Multiple R-squared:  0.1221, Adjusted R-squared:  0.08032
## F-statistic: 2.921 on 1 and 21 DF,  p-value: 0.1022
```

```
BetaT <- summary(lm.T)$coefficients[2, 1]
```

```
#Model for TRV
lm.TRV <- lm(TRV ~ SP500, data = as.data.frame(MyData))
summary(lm.TRV)
```

```
##
## Call:
## lm(formula = TRV ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.025969 -0.011764 -0.004341  0.014924  0.034865
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.001389   0.003895   0.357   0.725
## SP500       0.411462   0.268364   1.533   0.140
##
## Residual standard error: 0.01864 on 21 degrees of freedom
## Multiple R-squared:  0.1007, Adjusted R-squared:  0.05785
## F-statistic: 2.351 on 1 and 21 DF,  p-value: 0.1401
```

```
BetaTRV <- summary(lm.TRV)$coefficients[2, 1]
```

```
#Model for UTX
lm.UTX <- lm(UTX ~ SP500, data = as.data.frame(MyData))
summary(lm.UTX)
```

```
##
## Call:
## lm(formula = UTX ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.036825 -0.014469 -0.005822  0.015152  0.044403
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.002886   0.004537   0.636   0.532
## SP500       0.134830   0.312581   0.431   0.671
##
```

```
## Residual standard error: 0.02171 on 21 degrees of freedom
## Multiple R-squared: 0.008782, Adjusted R-squared: -0.03842
## F-statistic: 0.1861 on 1 and 21 DF, p-value: 0.6706
```

```
BetaUTX <- summary(lm.UTX)$coefficients[2, 1]
```

```
#Model for VZ
```

```
lm.VZ <- lm(VZ ~ SP500, data = as.data.frame(MyData))
summary(lm.VZ)
```

```
##
## Call:
## lm(formula = VZ ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.023881 -0.014426 -0.000661  0.009177  0.044751
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.0004706  0.0036771   0.128   0.899
## SP500        0.3599403  0.2533225   1.421   0.170
##
## Residual standard error: 0.0176 on 21 degrees of freedom
## Multiple R-squared: 0.08771, Adjusted R-squared: 0.04426
## F-statistic: 2.019 on 1 and 21 DF, p-value: 0.17
```

```
BetaVZ <- summary(lm.VZ)$coefficients[2, 1]
```

```
#Model for WMT
```

```
lm.WMT <- lm(WMT ~ SP500, data = as.data.frame(MyData))
summary(lm.WMT)
```

```
##
## Call:
## lm(formula = WMT ~ SP500, data = as.data.frame(MyData))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.064114 -0.007832  0.000849  0.012119  0.025335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.001897  0.004063  -0.467   0.645
## SP500        0.137241  0.279876   0.490   0.629
##
## Residual standard error: 0.01944 on 21 degrees of freedom
## Multiple R-squared: 0.01132, Adjusted R-squared: -0.03576
## F-statistic: 0.2405 on 1 and 21 DF, p-value: 0.629
```

```
BetaWMT <- summary(lm.WMT)$coefficients[2, 1]
```

```
#Model for XOM
```

```
lm.XOM <- lm(XOM ~ SP500, data = as.data.frame(MyData))  
summary(lm.XOM)
```

```
##  
## Call:  
## lm(formula = XOM ~ SP500, data = as.data.frame(MyData))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.060482 -0.015971  0.000493  0.015014  0.053415   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.0003439  0.0053718  -0.064   0.950      
## SP500         0.0539848  0.3700786   0.146   0.885      
##  
## Residual standard error: 0.02571 on 21 degrees of freedom  
## Multiple R-squared:  0.001012,    Adjusted R-squared:  -0.04656   
## F-statistic: 0.02128 on 1 and 21 DF,  p-value: 0.8854
```

```
BetaXOM <- summary(lm.XOM)$coefficients[2, 1]
```

```
# Display betas
```

```
paste("Beta of AA: ", BetaAA)
```

```
## [1] "Beta of AA:  0.306354776120703"
```

```
paste("Beta of AXP: ", BetaAXP)
```

```
## [1] "Beta of AXP:  0.801278586406885"
```

```
paste("Beta of BA: ", BetaBA)
```

```
## [1] "Beta of BA:  0.35814062155553"
```

```
paste("Beta of BAC: ", BetaBAC)
```

```
## [1] "Beta of BAC:  1.2565665507821"
```

```
paste("Beta of CAT: ", BetaCAT)
```

```
## [1] "Beta of CAT:  0.349837279026352"
```

```
paste("Beta of CSC0: ", BetaCSC0)
```

```
## [1] "Beta of CSC0: 0.204975853538089"
```

```
paste("Beta of CVX: ", BetaCVX)
```

```
## [1] "Beta of CVX: -0.155348031565196"
```

```
paste("Beta of DD: ", BetaDD)
```

```
## [1] "Beta of DD: 0.39969027038004"
```

```
paste("Beta of DIS: ", BetaDIS)
```

```
## [1] "Beta of DIS: 0.771826338820731"
```

```
paste("Beta of GE: ", BetaGE)
```

```
## [1] "Beta of GE: 0.715414496369996"
```

```
paste("Beta of HD: ", BetaHD)
```

```
## [1] "Beta of HD: 0.444162451791471"
```

```
paste("Beta of HPQ: ", BetaHPQ)
```

```
## [1] "Beta of HPQ: -0.235339447119812"
```

```
paste("Beta of IBM: ", BetaIBM)
```

```
## [1] "Beta of IBM: 0.283706884688943"
```

```
paste("Beta of INTC: ", BetaINTC)
```

```
## [1] "Beta of INTC: 0.1178111734686"
```

```
paste("Beta of JNJ: ", BetaJNJ)
```

```
## [1] "Beta of JNJ: -0.340508808878504"
```

```
paste("Beta of JPM: ", BetaJPM)
```

```
## [1] "Beta of JPM: 0.580051053760392"
```

```
paste("Beta of KO: ", BetaKO)
```

```
## [1] "Beta of KO: 0.175481246903899"
```

```
paste("Beta of KRFT: ", BetaKRFT)
```

```
## [1] "Beta of KRFT: 0.214370501404088"
```

```
paste("Beta of MCD: ", BetaMCD)
```

```
## [1] "Beta of MCD: 0.547377294139315"
```

```
paste("Beta of MMM: ", BetaMMM)
```

```
## [1] "Beta of MMM: 0.273591200231556"
```

```
paste("Beta of MRK: ", BetaMRK)
```

```
## [1] "Beta of MRK: 0.382451665861558"
```

```
paste("Beta of MSFT: ", BetaMSFT)
```

```
## [1] "Beta of MSFT: 0.110945264398497"
```

```
paste("Beta of PFE: ", BetaPFE)
```

```
## [1] "Beta of PFE: 0.479914759329808"
```

```
paste("Beta of PG: ", BetaPG)
```

```
## [1] "Beta of PG: 0.263308018325903"
```

```
paste("Beta of T: ", BetaT)
```

```
## [1] "Beta of T: 0.456480571124317"
```

```
paste("Beta of TRV: ", BetaTRV)
```

```
## [1] "Beta of TRV: 0.411461545045901"
```

```
paste("Beta of UTX: ", BetaUTX)
```

```
## [1] "Beta of UTX: 0.13482974300242"
```



```
paste("Beta of VZ: ", BetaVZ)
```

```
## [1] "Beta of VZ: 0.359940283752318"
```

```
paste("Beta of WMT: ", BetaWMT)
```

```
## [1] "Beta of WMT: 0.137241467534037"
```

```
paste("Beta of XOM: ", BetaXOM)
```

```
## [1] "Beta of XOM: 0.0539848182889091"
```

```
# Treasury data
```

```
treasury <- read_excel("/Users/doris/OneDrive/Documents/Treasury.xlsx")  
str(treasury)
```

```
## tibble [25 x 7] (S3: tbl_df/tbl/data.frame)  
## $ Date      : POSIXct[1:25], format: "2011-06-18" "2011-06-11" ...  
## $ Open      : num [1:25] 2.9 2.99 3 3.09 3.1 ...  
## $ High      : num [1:25] 3 3.1 3.06 3.1 3.16 ...  
## $ Low       : num [1:25] 2.89 2.89 2.92 2.95 3.06 ...  
## $ Close     : num [1:25] 2.91 2.94 2.97 3 3.06 ...  
## $ Adj Close : num [1:25] 2.91 2.94 2.97 3 3.06 ...  
## $ Volume    : chr [1:25] "-" "-" "-" "-" ...
```

```
colSums(is.na(treasury))
```

```
##      Date      Open      High      Low      Close Adj Close      Volume  
##      0         0         0         0         0         0         0
```

```
# Remove leading and trailing spaces from all column names
```

```
colnames(treasury) <- trimws(colnames(treasury))
```

```
# Check the column names again
```

```
colnames(treasury)
```

```
## [1] "Date"      "Open"      "High"      "Low"      "Close "  
## [6] "Adj Close " "Volume"
```

```
# Print the exact column names with quotation marks to see any hidden characters
```

```
print(colnames(treasury))
```

```
## [1] "Date"      "Open"      "High"      "Low"      "Close "  
## [6] "Adj Close " "Volume"
```

```
# Fix column names by completely redefining them
```

```
colnames(treasury) <- c("Date", "Open", "High", "Low", "Close", "AdjClose", "Volume")
```

```
# Now try to access the Close column
```

```
average_risk_free_rate <- mean(treasury$Close, na.rm = TRUE)
```

```
cat("The average risk-free rate is:", round(average_risk_free_rate, 4), "\n")
```

```

## The average risk-free rate is: 3.3133

# Define risk-free rate (using the average from your Treasury data)
Rf <- 3.3133/100 # Convert percentage to decimal

# Calculate the average market return (S&P500)
market_return <- mean(MyData[, "SP500"], na.rm = TRUE)

# Calculate excess market return
excess_market_return <- market_return - Rf

# Initialize a dataframe to store results
results <- data.frame(Stock = colnames(MyData)[-1], # Exclude S&P500
                      Beta = numeric(ncol(MyData) - 1),
                      ExpectedReturn = numeric(ncol(MyData) - 1))

# Calculate beta and expected returns for each stock
for (i in 2:ncol(MyData)) {
  # Calculate beta using covariance/variance
  stock_returns <- MyData[, i]
  market_returns <- MyData[, "SP500"]

  # Beta calculation
  beta <- cov(stock_returns, market_returns, use = "complete.obs") /
    var(market_returns, na.rm = TRUE)

  # Expected return using CAPM
  expected_return <- Rf + beta * excess_market_return

  # Store results
  stock_idx <- i - 1
  results$Beta[stock_idx] <- beta
  results$ExpectedReturn[stock_idx] <- expected_return * 100 # Convert to percentage
}

# Print results
print(results)

```

##	Stock	Beta	ExpectedReturn
## 1	AA	0.30635478	2.3272657
## 2	AXP	0.80127859	0.7343026
## 3	BA	0.35814062	2.1605876
## 4	BAC	1.25656655	-0.7310885
## 5	CAT	0.34983728	2.1873128
## 6	CSCO	0.20497585	2.6535642
## 7	CVX	-0.15534803	3.8133036
## 8	DD	0.39969027	2.0268558
## 9	DIS	0.77182634	0.8290977
## 10	GE	0.71541450	1.0106650
## 11	HD	0.44416245	1.8837175
## 12	HPQ	-0.23533945	4.0707642
## 13	IBM	0.28370688	2.4001603
## 14	INTC	0.11781117	2.9341126
## 15	JNJ	-0.34050881	4.4092626

##	16	JPM	0.58005105	1.4463461
##	17	KO	0.17548125	2.7484956
##	18	KRFT	0.21437050	2.6233265
##	19	MCD	0.54737729	1.5515100
##	20	MMM	0.27359120	2.4327186
##	21	MRK	0.38245167	2.0823400
##	22	MSFT	0.11094526	2.9562113
##	23	PFE	0.47991476	1.7686451
##	24	PG	0.26330802	2.4658161
##	25	T	0.45648057	1.8440704
##	26	TRV	0.41146155	1.9889688
##	27	UTX	0.13482974	2.8793366
##	28	VZ	0.35994028	2.1547952
##	29	WMT	0.13724147	2.8715742
##	30	XOM	0.05398482	3.1395443